

PAPER_379 — MUGE Dual-Model 7-System Numeric Comparison: Compressed vs Resonance

Session: 103 (Re-analysis pass — lines 2700–3100 read for first time)

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Source: grokshare11254865.txt, lines ~2700–3100 **Parent document:** "100. MUGE Compression cycle 3_11May2025.docx" (Grok full integration) **Session:** 103 (Re-analysis pass — lines 2700–3100 read for first time) **CP4 Class:** DualModelMUGEComparisonCalculator (CP4 #29)

1. Overview

When Grok integrated both MUGE documents ("100." Compressed + "200." Resonance), it produced a full **side-by-side numeric comparison** of both models for all 7 canonical astrophysical systems. This paper captures that comparison table, the dominant-term analysis for each system in each model, and the key theoretical insight about when the models converge.

This data was NOT captured in PAPER371 (resonance only) or PAPER372 (compressed only).

2. Full 7-System Numeric Comparison Table

System	Compressed MUGE g (m/s ²)	Resonance MUGE g (m/s ²)	Dominant Term (Compressed)	Dominant Term (Resonance)
Magnetar SGR 1745-2900	1.782 \times 10^{39}	1.773 \times 10^{-9}	Perturbation (M \cdot \delta \rho / \rho)	Fluid a_{fluid_freq}
Sagittarius A*	3.552 \times 10^{20}	4.105 \times 10^{29}	Fluid \rho_{fl} \cdot V \cdot g	Fluid a_{fluid_freq}
Tapestry of Blazing Starbirth	1.001 \times 10^{27}	1.001 \times 10^{27}	Fluid/Perturbation (tie)	Fluid a_{fluid_freq}
Westerlund 2	1.001 \times 10^{27}	1.001 \times 10^{27}	Fluid/Perturbation (tie)	Fluid a_{fluid_freq}
Pillars of Creation	2.001 \times 10^{26}	2.001 \times 10^{26}	Fluid	Fluid a_{fluid_freq}
Rings of Relativity	5.005 \times 10^{25}	5.005 \times 10^{25}	Fluid	Fluid a_{fluid_freq}
Student's Guide to the Universe	3.958 \times 10^{14}	3.958 \times 10^{14}	Fluid	Fluid a_{fluid_freq}

3. Key Observations

3.1 SGR 1745-2900 — Extreme Divergence (48 Orders)

The magnetar case shows the **largest discrepancy** between the two models:

Compressed MUGE: $g \approx 1.782 \times 10^3 \text{ m/s}^2$ (dominated by perturbation term)

Perturbation = $(M + M_{\text{DM}}) \cdot (\delta\rho/\rho + 3GM/r^3)$

= $(2.984 \times 10^3 + 0) \cdot (10 + 5.973 \times 10)$

$\approx 1.782 \times 10^3 \text{ m/s}^2$

Resonance MUGE: $g \approx 1.773 \times 10^3 \text{ m/s}^2$ (dominated by fluid term)

$a_{\text{fluid_freq}} = f_{\text{fluid}} \cdot E_{\text{vac,neb}} \cdot V_{\text{sys}} / E_{\text{vac,ISM}} / c$

= $1.269 \times 10^1 \times 7.09 \times 10^3 \times 4.189 \times 10^{12} / 7.09 \times 10^3 / 3 \times 10$

$\approx 1.773 \times 10^3 \text{ m/s}^2$

Physical interpretation: The compressed model's perturbation term is **unphysically large** for a magnetar (dense neutron star) — the $\delta\rho/\rho$ density perturbation applied to the full mass at neutron-star densities gives an unphysical result. The resonance model, which relies on vacuum energy density ratios and system volume, gives a physically plausible acceleration for a magnetar at scale.

This is evidence that the **Resonance MUGE is the more physically appropriate model for compact objects** (neutron stars, magnetars), while the Compressed MUGE is better suited to galactic/cosmological scales.

3.2 Sagittarius A — Sgr A Reversal (Resonance > Compressed by 9 Orders)

Compressed MUGE: $g \approx 3.552 \times 10^2 \text{ m/s}^2$ (dominated by fluid: $\rho_{\text{fl}} \cdot V \cdot g_{\text{local}}$)

Resonance MUGE: $g \approx 4.105 \times 10^2 \text{ m/s}^2$ (dominated by fluid: $a_{\text{fluid_freq}}$)

Both models are fluid-dominated for Sgr A*, but the resonance model gives a value 9 orders of magnitude **higher** — reflecting that the resonance model picks up the volume-scaled vacuum energy coupling ($E_{\text{vac,neb}} \cdot V_{\text{sys}}$) which is enormous for the SMBH volume $V_{\text{sys}} = 3.552 \times 10^{45} \text{ m}^3$.

3.3 Star-Forming Regions and Beyond — Model Convergence

For the 5 remaining systems (Tapestry, Westerlund, Pillars, Rings, Student's Guide), **both models converge to the same value**. This is the empirical confirmation of the Cohesive UQFF principle (PAPER_378): in the star-forming/diffuse/large-volume regime, the resonance and compressed models give the same result — the fluid dynamics term dominates in BOTH models and is governed by the same physical inputs.

4. System Parameters (Full Reference)

System 1: Magnetar SGR 1745-2900

$M = 2.984 \times 10^3 \text{ kg}$, $r = 10 \text{ m}$, $t = 3.799 \times 10^1 \text{ s}$, $z = 0.0009$

$B = 10^4 \text{ T}$, $B_{\text{crit}} = 10^{11} \text{ T}$ ($B/B_{\text{crit}} = 0.1 \rightarrow 0.9$ factor)

$\rho_{\text{fluid}} = 10^1 \text{ kg/m}^3$, $V = 4.189 \times 10^{12} \text{ m}^3$, $g_{\text{local}} = 10 \text{ m/s}^2$

$M_{\text{DM}} = 0$, $\delta\rho/\rho = 10$

Resonance: $I=10^{21} \text{ A}$, $A=3.142 \times 10^2 \text{ m}^2$, $\omega=10^3 \text{ rad/s}$, $\omega=-10^3 \text{ rad/s}$

$v_{\text{exp}} = 10^3 \text{ m/s}$, $f_{\text{fluid}} = 1.269 \times 10^1 \text{ Hz}$

System 2: Sagittarius A*

$M = 8.155 \times 10^3 \text{ kg}$, $r = 10^{12} \text{ m}$, $t = 3.786 \times 10^1 \text{ s}$, $z = 0.0009$

$B = 10^4 \text{ T}$, $B_{\text{crit}} = 10^{11} \text{ T}$ ($B/B_{\text{crit}} = 0.1 \rightarrow 0.9$ factor)

$\rho_{\text{fluid}} = 10^2 \text{ kg/m}^3$, $V = 3.552 \times 10^{45} \text{ m}^3$, $g_{\text{local}} = 10 \text{ m/s}^2$

$M_{\text{DM}} = 10^3 \text{ kg}$, $\delta\rho/\rho = 10$

Resonance: $I=10^{23} \text{ A}$, $A=2.813 \times 10^3 \text{ m}^2$, $\omega=10^3 \text{ rad/s}$, $\omega=-10^3 \text{ rad/s}$

$v_{\text{exp}} = 5 \times 10^3 \text{ m/s}$, $f_{\text{fluid}} = 3.465 \times 10^1 \text{ Hz}$

Systems 3–7 (Condensed)

Tapestry: $M=10^4 M_\odot$, $r=10 \text{ pc}$, $V=10^{23} \text{ m}^3$, $f_{\text{fluid}}=10^{12} \text{ Hz}$
Westerlund 2: Same as Tapestry (young cluster, stellar winds)
Pillars of Creation: $M=10^2 M_\odot$, $r=1 \text{ ly}$, $V=10^{22} \text{ m}^3$, $f_{\text{fluid}}=10^{11} \text{ Hz}$
Rings of Relativity: $M=10^4 M_\odot$, $r=10 \text{ pc}$, $V=10^{23} \text{ m}^3$, $f_{\text{fluid}}=10^{12} \text{ Hz}$
Student's Guide: $M \approx 10^{23} \text{ kg}$ (universe), $r=10^2 \text{ m}$, $V=10^{23} \text{ m}^3$, $f_{\text{fluid}}=10^{11} \text{ Hz}$

5. The Fluid Dynamics Universality Principle

A key result from this comparison: **in EVERY system in BOTH models, the fluid dynamics term ultimately dominates** (with the exception of the SGR1745 compressed case where the perturbation term wins).

Fluid dynamics universality:

For Resonance MUGE: $a_{\text{fluid_freq}} = f_{\text{fluid}} \cdot E_{\text{vac,neb}} \cdot V_{\text{sys}} / E_{\text{vac,ISM}} / c$
For Compressed MUGE: $\text{fluid_term} = \rho_{\text{fluid}} \cdot V_{\text{sys}} \cdot g_{\text{local}}$

Both terms scale as $\propto V_{\text{sys}}$ — larger systems have proportionally larger fluid acceleration. This is consistent with Navier-Stokes turbulence being the governing dynamics at astrophysical scales (connecting to PAPER_369).

6. Unit Test Reference Values

Derived from the Grok computation (validated against Grok's work-shown derivations):

System	Compressed g	Resonance g	Notes
SGR 1745-2900	1.782×10^3	1.773×10^3	Resonance = physical; compressed unphysical
Sgr A*	3.552×10^{20}	4.105×10^2	Both fluid-dominated
Tapestry	1.001×10^2	1.001×10^2	Convergent
Westerlund	1.001×10^2	1.001×10^2	Convergent
Pillars	2.001×10^2	2.001×10^2	Convergent
Rings	5.005×10^2	5.005×10^2	Convergent
Student's Guide	3.958×10^1	3.958×10^1	Convergent

7. Implementation

C++ (grokshare11254865.txt, dual-model main()):

```
for (const auto& sys : muge_systems) {
double compressed_g = compute_compressed_MUGE(sys);
double resonance_g = compute_resonance_MUGE(sys, res_params);
std::cout << "Compressed MUGE g for " << sys.name << ": " << compressed_g << " m/s2\n";
std::cout << "Resonance MUGE g for " << sys.name << ": " << resonance_g << " m/s2\n";
}
```

Python CP4 class: DualModelMUGEComparisonCalculator (CP4 class #29)

8. CP4 Class

Class: `DualModelMUGECComparisonCalculator` **Category:** MUGE Validation / Comparison **Key method:** `compute(dataset)` — takes 7-system catalog, returns compressed vs resonance table **References:** PAPER371, PAPER372, PAPER_378

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